FINTECHATHON 2026 · INTERNATIONAL TRACK · TOPIC F · SUBMISSION ITEM 4

Shouhu 守护 — Source Code

README, GitHub repository link, deployment instructions and test cases

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Repository https://github.com/SeerieTheGreat/AIAgent (public) · 20 October 2026

At a glance

Required by the rules	Where it is
GitHub repository link	https://github.com/SeerieTheGreat/AIAgent · public · branch `main`
Source code	The repository above; the submission zip holds an exact copy of it in `4_Source_Code/AIAgent/`
README	`README.md`, the repository's front page on GitHub
Deployment instructions	README sections "Quick start", "Interactive use", "Browser interface" and "Deploy your own copy"; summarised in section 2 below
Test cases	`docs/test_cases.md`: 86 cases (12 contract tests, 58 Python tests, 8 scripted tasks, 8 attacks), each with what it checks; results in `docs/test_results/`

Paths are relative to the repository root (in the submission zip, `4_Source_Code/AIAgent/`). Everything below was checked on a fresh copy of the repository on 20 October 2026: every README step ran without changes, and every test passed (section 3). This guide is also in the repository as `docs/Shouhu_Source_Code_Guide.pdf`.

1. The repository

Folder	Contents
`contracts/`	Solidity smart contracts: `PolicyWallet.sol` (owner and agent roles, per-transfer cap, rolling 24-hour cap, allowlist, pause, one-time approvals, intent ids) and `TestUSDC.sol`; Foundry tests in `contracts/test/`
`shouhu/`	Python package: agent loop, five tools, policy engine, screening, audit log, chain adapter, rules-only assistant, owner tools, web app and owner console (`shouhu/static/` holds the pages), proof and ledger, judge mode, red team, deployment
`tests/`	58 Python tests: policy, screening, money, agent loop, end to end on a local chain, rules-only assistant, web app
`docs/`	Technical documentation (PDF and Markdown), test cases, security self-assessment, demo guide
`evidence/`	Audit logs, red-team results, exported runs with transaction hashes (including all 20 live Sepolia payments)
`deployments/`, `scripts/`	Deployed addresses per chain; Sourcify verification script

- **Languages:** Solidity 0.8 (contracts), Python 3.12 (agent and services), HTML, CSS and JavaScript (pages, no external scripts). Python dependencies are pinned in `uv.lock`; the compiled contracts are included, so no Solidity compiler is needed to run the agent.

- **No secrets in the package:** `.env` and `.owner.env` are git-ignored and not included. Each user generates new test keys with one command (section 2). tUSDC is a test token with no value.
- **Rights:** all rights reserved by the Shouhu team; see the README.

2. Deployment instructions

2.1 Requirements

macOS or Linux with uv (Python 3.12 is installed by uv) and Foundry (for the local chain and the contract tests). An Anthropic API key enables the AI model; without one, the offline judge mode, the rules-only red team and the browser app's rules-only mode all work.

```
curl -LsSf https://astral.sh/uv/install.sh | sh          # uv, once
curl -L https://foundry.paradigm.xyz | bash && foundryup  # Foundry, once
git clone https://github.com/SeerieTheGreat/AIAgent.git && cd AIAgent
uv sync                                                   # Python environment
uv run python -m shouhu.keys                              # new test keys (.env, .owner.env)
```

2.2 Judge mode on a local chain (no faucet, no accounts)

Each command starts its own local chain, deploys the contracts and exports an evidence bundle. The AI-model runs need `ANTHROPIC_API_KEY` in `.env`.

```
uv run shouhu-demo --chain anvil --offline   # 8 scripted tasks, no model
uv run shouhu-demo --chain anvil             # the same with the AI model
uv run shouhu-redteam --chain anvil --rules  # 8 attacks, rules-only assistant
uv run shouhu-redteam --chain anvil          # 8 attacks, AI model
```

2.3 The browser app

```
uv run shouhu-web --chain anvil    # fresh local chain (or --chain sepolia)
# student chat   http://127.0.0.1:8787/
# owner console  http://127.0.0.1:8788/   (separate process, owner key)
```

2.4 Your own deployment on Ethereum Sepolia

First send about 0.05 Sepolia test ether to the owner address in `ADDRESSES.md` (written by the key command), for example from the Google Cloud or Chainlink faucet. `shouhu-deploy` then deploys TestUSDC and PolicyWallet (caps 500 and 1,000 tUSDC), mints 10,000 tUSDC into the wallet, allowlists the five contacts and sends the agent 0.01 ETH for gas.

```
uv run shouhu-deploy --chain sepolia      # contracts, test money, contacts
bash scripts/verify_sourcify.sh sepolia   # optional: source on Sourcify
uv run shouhu-web --chain sepolia
```

2.5 The team's live deployment

Item	Value
PolicyWallet	`0xb9D959da30680e86E9bF0A287Abee354033a0525` (Sourcify exact match)
TestUSDC	`0x6b2f3F61DB4e45D07453D58dF2F4eB27C5284931` (Sourcify exact match)
Network and block	Ethereum Sepolia (chain id 11155111), deployed at block 11,926,706

Item	Value
Check without keys	`uv run shouhu-try --chain sepolia mama 600` (the contract answers REJECTED) and `uv run shouhu-verify --chain sepolia <hash>`

3. Test cases and results

Suite	Cases	Command	Result on the fresh copy
Smart-contract tests (Foundry)	12	`cd contracts && forge install foundry-rs/forge-std@v1.17.0 --no-git && forge test`	12 passed
Python unit and integration tests	58	`uv run pytest`	58 passed
Scripted end-to-end tasks	8	`uv run shouhu-demo --chain anvil --offline`	8 of 8 as expected
Red-team attacks	8	`uv run shouhu-redteam --chain anvil --rules`	8 of 8 passed

Every case is listed with what it checks in `docs/test_cases.md`. The attacks were also run against the AI model: 8 of 8 passed (`evidence/redteam_results.md`).

Fresh-copy run. On 20 October 2026 at 16:38 UTC+8, commit `05785df` (the last change to the code; later commits add only this guide and the logs) was exported from the repository into an empty folder and every README step was run there (macOS 26.4, uv 0.13.1, Python 3.12.15, Foundry 1.8.5). The logs are in `docs/test_results/`:

Log	What it shows
`01_install_uv_sync.txt`	The Python environment installs from `uv.lock`
`02_python_tests_58_passed.txt`	All 58 Python tests pass, 22 of them on a local Anvil chain
`03_contract_test_library_forge_install.txt`	The Foundry test library installs
`04_contract_tests_12_passed.txt`	All 12 contract tests pass
`05_judge_mode_offline_8_tasks.txt`	Judge mode: the 8 scripted tasks on a fresh local chain; final balance 9,170 tUSDC as expected
`06_red_team_rules_only_8_of_8.txt`	8 of 8 attacks produce no payment and leak nothing
`07_deploy_to_local_chain.txt`	`shouhu-deploy` deploys both contracts, mints the test money and allowlists the contacts
`08_contract_dry_run_sepolia_600_rejected.txt`	The live contract rejects 600 tUSDC (over the 500 cap), with no keys
`09_contract_dry_run_sepolia_5_allowed.txt`	The live contract allows 5 tUSDC to Mama (dry run, nothing sent)
`10_verify_live_transaction_sepolia.txt`	A live Tier 2 payment re-read from Sepolia and linked to the student's words and the policy decision in the committed audit log